

In [45]:

```
#Day 5
#Python tuple
#sequence of immutable Python objects.
```

In [46]:

```
#declare the tuple
tuple1 = (1,2,3,4,5)
tuple2 = (4,5,6,7,8)
tuple3 = ()

print(tuple1)
print(tuple2)
print(tuple3)
```

```
(1, 2, 3, 4, 5)
(4, 5, 6, 7, 8)
()
```

In [47]:

```
tuple[1] = 34
```

```
-----
-
TypeError                                Traceback (most recent call last)
<ipython-input-47-44ec8c35a790> in <module>
----> 1 tuple[1] = 34
```

TypeError: 'type' object does not support item assignment

In [48]:

```
print(tuple1[1])
print(tuple1[0:])
print(tuple1[:3])
print(tuple1[:])
```

```
2
(1, 2, 3, 4, 5)
(1, 2, 3)
(1, 2, 3, 4, 5)
```

In [51]:

```
print(tuple1[-1])
print(tuple1[-3:-1])
print(tuple1[-3:])
```

```
5
(3, 4)
(3, 4, 5)
```

In [52]:

```
print(tuple1[8])
```

```
-----  
-  
IndexError                                Traceback (most recent call las  
t)  
<ipython-input-52-1db2bc9c55d7> in <module>  
----> 1 print(tuple1[8])
```

IndexError: tuple index out of range

In [55]:

```
print(tuple1)  
print(tuple1[0:len(tuple1):2])
```

```
(1, 2, 3, 4, 5)  
(1, 3, 5)
```

In [56]:

```
#immutable objects  
del tuple1[2]
```

```
-----  
-  
TypeError                                Traceback (most recent call las  
t)  
<ipython-input-56-489f44525901> in <module>  
    1 #immutable objects  
----> 2 del tuple1[2]
```

TypeError: 'tuple' object doesn't support item deletion

In [57]:

```
del tuple1
```

In [58]:

```
print(tuple1)
```

```
-----  
-  
NameError                                Traceback (most recent call las  
t)  
<ipython-input-58-e27b56724cb0> in <module>  
----> 1 print(tuple1)
```

NameError: name 'tuple1' is not defined

In [60]:

```
tuple3 = tuple2*2    #Repetition
print(tuple2)
print(tuple3)
```

```
(4, 5, 6, 7, 8)
(4, 5, 6, 7, 8, 4, 5, 6, 7, 8)
```

In [62]:

```
tuple4 = tuple2 + tuple3 + (56,67)  #Concatenation
print(tuple4)
```

```
(4, 5, 6, 7, 8, 4, 5, 6, 7, 8, 4, 5, 6, 7, 8, 56, 67)
```

In [63]:

```
print(56 in tuple4)  #Membership
```

```
True
```

In [66]:

```
num = int(input("Enter the number:: "))
count = 0
for var in tuple4:    #iteration
    if(num == var):
        count += 1
print(num , " occurs " , count , "times")
```

```
Enter the number:: 56
56 occurs 1 times
```

In [68]:

```
print(len(tuple4))
print("Max = " , max(tuple4))
print("Min = " , min(tuple4))
```

```
17
Max = 67
Min = 4
```

In [69]:

```
myTuple = (4, 2, 5,7, 1, 8, 9)
print(myTuple)
```

```
(4, 2, 5, 7, 1, 8, 9)
```

In [78]:

```
sortedlist1 = sorted(list(myTuple))
print("Sorted List = " , sortedlist1)
print("Max 2 elements :: " , tuple(sortedlist1[-2:]))
print("Min 2 elements :: " , tuple(sortedlist1[0:2]))
restuple = tuple(sortedlist1[0:2]) + tuple(sortedlist1[-2:])
print(restuple)
```

Sorted List = [1, 2, 4, 5, 7, 8, 9]
 Max 2 elements :: (8, 9)
 Min 2 elements :: (1, 2)
 (1, 2, 8, 9)

In [83]:

```
print("Sorted List = " , sortedlist1)
k = int(input("Enter the req value:: "))

print("Max k elements :: " , tuple(sortedlist1[-k:]))
print("Min k elements :: " , tuple(sortedlist1[0:k]))
restuple = tuple(sortedlist1[0:k]) + tuple(sortedlist1[-k:])
print(restuple)
```

Sorted List = [1, 2, 4, 5, 7, 8, 9]
 Enter the req value:: 3
 Max k elements :: (7, 8, 9)
 Min k elements :: (1, 2, 4)
 (1, 2, 4, 7, 8, 9)

In [97]:

```
#list of tuples
listoftuples1 = [("Rohan", 34,"Account"),("Mohan", 76,"MBA"),("Joe", 25,"HR")]
print(listoftuples1)
```

[('Rohan', 34, 'Account'), ('Mohan', 76, 'MBA'), ('Joe', 25, 'HR')]

In [98]:

```
#iterate on list of tuples
for eachtuple in listoftuples1:
    for eachelement in eachtuple:
        print(eachelement, end = " ")
    print(' ')
```

Rohan 34 Account
 Mohan 76 MBA
 Joe 25 HR

In [102]:

```
#sort the listoftuples (based on name)
listoftuples2 = [("Rohan", 34, "Account"), ("Mohan", 76, "MBA"), ("Joe", 25, "HR")]
print(listoftuples2)
listoftuples2.sort(key = lambda x: x[-1]) #Department
print(listoftuples2)

listoftuples2.sort(key = lambda x: x[-2]) #age
print(listoftuples2)

listoftuples2.sort(key = lambda x: x[-3]) #name
print(listoftuples2)
```

```
[('Rohan', 34, 'Account'), ('Mohan', 76, 'MBA'), ('Joe', 25, 'HR')]
[('Rohan', 34, 'Account'), ('Joe', 25, 'HR'), ('Mohan', 76, 'MBA')]
[('Joe', 25, 'HR'), ('Rohan', 34, 'Account'), ('Mohan', 76, 'MBA')]
[('Joe', 25, 'HR'), ('Mohan', 76, 'MBA'), ('Rohan', 34, 'Account')]
```

In [113]:

```
tupList = [(5, 7), (20, 13), (45, 2), (12, 4)]
print("Before Sorting : " , tupList)

listLen = len(tupList)
for i in range(0, listLen):
    for j in range(0, listLen - i - 1):
        if(tupList[j][-2] > tupList[j + 1][-2]): #change -2 to -1 , if need to sort
                                                    #based on 2nd subelement

            swap = tupList[j]
            tupList[j] = tupList[j + 1]
            tupList[j + 1] = swap
print("After Sorting : " , tupList)
```

```
Before Sorting : [(5, 7), (20, 13), (45, 2), (12, 4)]
After Sorting : [(5, 7), (12, 4), (20, 13), (45, 2)]
```

In [119]:

```
#remove the duplicate tuple irrespective of the order
tupList = [(3, 1), (5, 8), (1, 3), (8, 4), (2, 9)]

# to sort the elements of each tuple in the list (way1)
sortedtuple = [tuple(sorted(val)) for val in tupList]
print(sortedtuple)

# to sort the elements of each tuple in the list (way2)
sortedtuple1 = []
for val in tupList:
    sortedtuple1.append(tuple(sorted(val)))
print(sortedtuple1)

#remove the duplicate
req = set(sortedtuple)
req1 = list(set(sortedtuple1))

print("=====")
print(req)
print(req1)
```

```
[(1, 3), (5, 8), (1, 3), (4, 8), (2, 9)]
[(1, 3), (5, 8), (1, 3), (4, 8), (2, 9)]
=====
{(1, 3), (2, 9), (5, 8), (4, 8)}
[(1, 3), (2, 9), (5, 8), (4, 8)]
```

In [131]:

```
tuplist5 = [("Sanjay", 10), ("Joe", 5), ("Madaan", 10), ("Roy", 8)]
print(tuplist5)

#concatenate the string values of the tuples with maximum associated values.
# Sanjay Madaan
maxvalue = -1
str1 = ""
for i in range(0, len(tuplist5)):
    if tuplist5[i][1] > maxvalue:
        maxvalue = tuplist5[i][1]
print("Max value = ", maxvalue)

for i in range(0, len(tuplist5)):
    if tuplist5[i][1] == maxvalue:
        str1 += tuplist5[i][0] + " "
print(str1)
```

```
[('Sanjay', 10), ('Joe', 5), ('Madaan', 10), ('Roy', 8)]
Max value = 10
Sanjay Madaan
```

In [139]:

```
#concatenate the string values of the tuples with maximum associated values.  
# Sanjay Madaan
```

```
from operator import itemgetter
```

```
print(tuplist5)
```

```
maxvalue = max(tuplist5, key=itemgetter(1))[1]
```

```
print(maxvalue)
```

```
reqstr = ' '.join(key for key, ele in tuplist5 if ele == maxvalue)
```

```
print(reqstr)
```

```
[('Sanjay', 10), ('Joe', 5), ('Madaan', 10), ('Roy', 8)]
```

```
10
```

```
Sanjay Madaan
```

In [146]:

```
signlist = [(-1, 3), (5, -8), (-1, 3), (4, -8), (2, 9)]
```

```
print(signlist)
```

```
#change the signs of each element in each tuple in the list
```

```
reqlist = []
```

```
for tup in signlist:
```

```
    reqlist.append(    (-1*tup[0], -1*tup[1])    )
```

```
print(reqlist)
```

```
reqlist1 = [(-1*tup[0], -1*tup[1]) for tup in signlist]
```

```
print(reqlist1)
```

```
[(-1, 3), (5, -8), (-1, 3), (4, -8), (2, 9)]
```

```
[(1, -3), (-5, 8), (1, -3), (-4, 8), (-2, -9)]
```

```
[(1, -3), (-5, 8), (1, -3), (-4, 8), (-2, -9)]
```

In []: